

Automated UCC-128 Label Printing by Box Number Scan

Functional & Technical Specification

System Acumatica ERP — Asgard Label Integration

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1. Executive Overview

This specification defines the functional and technical requirements for implementing an automated **UCC-128 label printing process** within Acumatica ERP, leveraging the Asgard Label integration layer.

The solution enables warehouse and shipping personnel to **trigger customer-specific UCC-128 label generation through a single box number scan** — eliminating manual lookup steps, reducing label errors, and accelerating throughput at the point of packing and dispatch.

2. Objectives

The following outcomes are targeted by this development:

- Automate the end-to-end printing of **UCC-128 labels** with no manual intervention beyond the initial scan.
 - Enable label printing to be triggered exclusively by **scanning a box number**, ensuring speed and accuracy at the operator level.
 - Support **customer-specific label configurations** through a structured Group Label model.
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3. Functional Requirements

3.1 Automated Label Printing

The system shall automatically print a UCC-128 label upon receipt of a valid box number scan. No additional user interaction is required after the scan event.

Workflow Summary

1. User navigates to the **UCC-128 Auto Print** screen.
2. User scans or manually enters a **Box Number** into the designated field.
3. System validates the box number against shipment package records.
4. System retrieves and auto-populates the associated **Shipment Number, Customer, and Box information**.
5. System dispatches the print job for the applicable **Group Label** to the configured printer.
6. System displays a confirmation message and clears the field for the next scan.

4. Label Configuration

4.1 Customer-Specific Label Model

UCC-128 label formats vary by customer and must be handled dynamically. The solution shall implement the following structure:

Group Label Architecture

A **Group Label** acts as a container that holds multiple UCC-128 label models. Each model within the group corresponds to a specific customer configuration. At print time, the system resolves the correct label model based on the customer associated with the scanned box number.

This architecture allows new customer label templates to be added without structural changes to the automation logic.

5. Custom Screen Specification

A dedicated custom screen shall be developed to support the scanning workflow and automated printing trigger.

Attribute	Detail
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Screen Name	UCC-128 Auto Print
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Screen ID	To be assigned
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Purpose	Scan box numbers and trigger automated, customer-specific label printing
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5.1 Screen Layout

Field	Type	Description
Box Number	Input / Scannable	Primary trigger field; initiates the lookup and print sequence on entry
Shipment Number	Read-only	Auto-populated following a successful box number lookup
Customer	Display only	Resolved from the shipment record; confirms the label model to be applied
Status Message	System feedback	Displays success confirmation or descriptive error messages

5.2 Screen Behaviour

Upon entry of a box number, the screen shall execute the following sequence automatically:

1. Validate the scanned box number against shipment package records.
2. Retrieve the associated shipment and customer records.
3. Resolve and dispatch the correct customer-specific label to the configured printer.
4. Display a success confirmation message.
5. Clear the Box Number field to prepare for the next scan.

All steps occur without additional user input. In the event of a validation failure, the screen displays the appropriate error message and awaits the next scan.

6. Box Number Validation Rules

6.1 Uniqueness Requirement

Box numbers are treated as **globally unique** across all shipment records within the system. This uniqueness is a prerequisite for the single-scan, no-confirmation workflow.

6.2 Validation Logic

Upon receiving a box number, the system shall query all shipment package records and apply the following logic:

Condition	System Action
Exactly one matching record found	Proceed with shipment retrieval and label printing
No matching record found	Halt process; display error message
Multiple matching records found	Halt process; display error message

6.3 Error Messages

Condition	Message Displayed
No match found	<i>"Box number not found."</i>
Duplicate records detected	<i>"Duplicate box numbers detected. Contact your system administrator."</i>

7. Automation Logic

7.1 Trigger Event

The automation is initiated exclusively by the **entry or scan of a Box Number** in the UCC-128 Auto Print screen.

7.2 Processing Sequence

1. Box Number received (scan or manual entry)
2. Query shipment package records for matching Box Number
3. Validate record uniqueness (see Section 6.2)
4. Retrieve: ShipmentNbr, CustomerID
5. Resolve applicable Group Label model for CustomerID
6. Dispatch print job to Asgard Label-configured printer
7. Display confirmation message; clear input field

8. Assumptions & Dependencies

The following conditions are assumed to be in place prior to deployment:

- Box numbers are unique across all shipment records in the Acumatica instance.
- UCC-128 label templates for all relevant customers are already defined and configured within the system.
- Physical printers are provisioned and mapped within the Asgard Label printer configuration.
- Users operating the screen have the appropriate Acumatica role permissions to access shipping and label functions.

9. Deliverables

The following components shall be delivered as part of this engagement:

Deliverable	Description
UCC-128 Auto Print Screen	Custom Acumatica screen with scanning input and auto-populated display fields
Box Number Scanning Automation	Event-driven logic to trigger lookups and print jobs on scan
Printer Integration	Automated print job dispatch via Asgard Label configuration
Validation & Error Handling	Full input validation with user-facing error messaging per Section 6